

Total No. of Questions : 8]

SEAT No. :

PC2345

[6354]-461

[Total No. of Pages :2

B.E. (Civil)

**INTEGRATED WATER RESOURCE
PLANNING AND MANAGEMENT**

(2019 Pattern) (Semester- VII) (Elective - III) (401003 C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Use of logarithmic table, slide rule and electronic pocket calculator are allowed.
- 5) Assume suitable data if necessary, stating it clearly.

Q1) a) What is ground water pollution? How it can be prevented & controlled? **[6]**

b) What are various cost effective water methods of water quality monitoring for basins? **[6]**

c) Explain benefits of Industrial waste water treatment? **[5]**

OR

Q2) a) What is Environmental Impact Assessment (EIA)? Explain the steps involved in EIA? **[6]**

b) State the Objectives and Functions of CPCB? **[6]**

c) State polluter pays principle? **[5]**

Q3) a) What is water scarcity? Explain its effects? **[6]**

b) Write a note on importance of water to the Indian economy? **[6]**

c) Write a note on cash flow diagram and discount rate? **[6]**

OR

Q4) a) State seven steps of water planning process. **[6]**

b) Explain sustainability principles for water management. **[6]**

c) State and explain the measures of preventing water scarcity. **[6]**

P.T.O.

- Q5) a)** Explain municipal corporation laws regarding water supply and drainage. [6]
b) Write a short note on Water allocation priorities. [6]
c) State the role and functions of CWC. [6]

OR

- Q6) a)** State four principles related water sector reform Dublin Principles (1992). [6]
b) Explain importance of arbitration in IWRPM. [6]
c) Write a short note on Inter-Basin Water Transfer. [6]

- Q7) a)** Explain the role of dam in flood control and its importance in IWRPM. [6]
b) Write a short note on flood forecasting and disaster relief. [6]
c) Explain application of soft computing in flood control. [5]

OR

- Q8) a)** Explain the role of dam in power generation and its importance in IWRPM. [6]
b) Compute the volume of water required to generate 4750 kW-hr of electric energy for one day if the average head is 95.0 m and efficiency of the power plant is 0.85. ($T = 24$ hr). [6]
c) State applications of QGIS in IWRPM. [5]

